

Nexora: Urban Flow Analytics

End-to-End Machine Learning Platform for Taxi Operations

SLIIT Codefest Datathon 2026 | Technical Report

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Final Submission Notebook: Nexora_FinalNotebook.ipynb

[Live Production System](#)

Code Repository: <https://github.com/praveen-madawalage/nexora-datathon2026.git>

Demo Video : <https://youtu.be/74t6G71aNkQ>

Executive Summary

Urban taxi networks represent the circulatory system of major metropolitan areas, generating millions of trip transactions daily. Raw operational records are notoriously fraught with sensor anomalies, billing artifacts, spatial ambiguities, and temporal volatility. Team Nexora designed, engineered, and deployed an end-to-end, production-grade machine learning platform trained across 44,459,188 cleaned taxi records spanning April 2025 through March 2026.

Our platform delivers an integrated operational suite addressing the competition's core objectives and bonus tracks:

- **Zero-Leakage Upfront Pricing Engine (Task 2.1):** Locks in guaranteed pre-trip fares using LightGBM, achieving a test Mean Absolute Error (MAE) of \$3.61 and an R^2 of 0.802 on the out-of-time March 2026 split.
- **On-Time Arrival & Duration Estimator (Task 2.2):** Predicts trip durations down to the minute, delivering 80.80% accuracy within ± 5 minutes and 93.54% within ± 10 minutes (Median Error: 2.11 min).
- **Fleet Dispatch Demand Forecaster (Task 3.1):** Regularized autoregressive time-series models across high-volume taxi zones, forecasting 24 to 72 hour pickup surges with an R^2 of 0.941 (overall test set) and 0.945 at the 72-hour full-month horizon.
- **Spatial-Temporal Hotspot & OD Flow Clustering (Task 3.2):** Discovers 5 functional mobility archetypes across 251 active zones via K-Means and Silhouette optimization, mapping diurnal commuter and nocturnal entertainment shifts.
- **AI Mobility Assistant (Bonus Track 5):** An intelligent, conversational question-answering agent powered by Google Gemini Flash and DuckDB, executing natural-language-to-SQL queries in approximately 8 ms behind dual-layer AST safety guardrails.
- **Executive Business Intelligence Dashboard (Bonus Track 6):** A decision-maker cockpit with an interactive Fleet Capacity Shift Simulator demonstrating that a 15% reallocation of idle dead-hour vehicles generates +\$5.2M in annual revenue.

1. Data Preprocessing & Feature Engineering

Raw Taxi Data (44.5M+ rows) -> DuckDB Cleaning SQL -> Model-Clean Tier (44,459,188 rows)
-> Strict Chronological Splits

1.1 Data Quality Profiling & Anomaly Quantification

Raw trip data contained significant operational and sensor anomalies. We built an automated, single-pass DuckDB SQL cleaning pipeline that categorizes every record into either valid retention, imputation, or domain-justified filtering:

Anomaly Type / Column	Filter / Impute Rule	Justification	Rows Impacted	% of Corpus
Negative Fares (base_fare < 0)	Filter (DROP)	Reversals, chargebacks, and dispute transactions cannot represent positive transport costs.	64,112	0.144%
Unrealistic Distance / Fare (distance = 0 & base_fare > 0)	Conditional Retention	>98% of these records were non-standard rates (Rate Code 2: JFK Flat, Rate Code 5: Negotiated corporate fares). Retained with rate-class interaction flags.	170,031	0.382%
Zero/Missing Passengers (rider_count NULL or 0)	Impute (rider_count = 1) + Flag	Driver meter activation without manual entry. Single-passenger assumption represents	8,271,402	22.08%

Anomaly Type / Column	Filter / Impute Rule	Justification	Rows Impacted	% of Corpus
		>72% of known trips.		
Missing Rate Class (rate_class_id NULL)	Impute (rate_class_id = 99) + Flag	Unclassified fare structure; assigned unknown category to preserve spatial coordinates and mileage.	8,072,119	18.15%
Temporal Inversion (dropoff <= pickup)	Filter (DROP)	Data corruption or negative duration; physically impossible journeys.	2,845	0.006%
Excess Speed (distance/duration > 100 mph)	Filter (DROP)	GPS telemetry loss or corrupted distance/timestamp pairing produces physically impossible average speeds.	14,892	0.033%
Stuck Meter (speed < 1 mph and duration over a separate threshold, distance > 0)	Filter (DROP)	Meter left running while parked or idling. Evaluated as its own rule alongside the excess-speed filter.	included above	included above
Fee Imputations (Congestion, Airport)	Impute (COALESCE(fee, 0.0))	Missing fees indicate zero charge for	12,405,268	27.90%

Anomaly Type / Column	Filter / Impute Rule	Justification	Rows Impacted	% of Corpus
		non-applicable zones rather than missing values.		

Final Cleaned Dataset Retention: exactly 44,459,188 valid operational records were retained in the model_clean gold tier with zero NULLs across all 24 schema columns.

1.2 Chronological Split Strategy (Zero Temporal Leakage)

To mirror real-world production deployment, we enforced a strict chronological partition without overlap:

- Training Set (April 1, 2025 to January 31, 2026): 37,459,831 records (84.26%).
- Validation Set (February 1, 2026 to February 28, 2026): 3,221,802 records (7.25%).
- Held-Out Test Set (March 1, 2026 to March 31, 2026): 3,777,555 records (8.49%).

$\max(\text{Train timestamp}) < \min(\text{Validation timestamp}) < \min(\text{Test timestamp})$. Random k-fold splits were rejected because they introduce temporal look-ahead leakage, artificially inflating validation metrics.

1.3 Feature Engineering: Zero-Leakage Pre-Trip Formulation

For upfront fare and arrival estimation, we engineered 30 features using only information available prior to trip departure:

- **Temporal & Cyclical Features:** pickup hour and day-of-week encoded with sine/cosine transformations; binary flags for rush hours (07:00 to 09:59, 16:00 to 19:59), late-night leisure windows (23:00 to 04:00), and weekend indicators.
- **Spatial Topology & Crossing Features:** origin/destination Zone IDs (high-cardinality categorical), is_same_zone, is_interborough, and is_manhattan_intra.
- **Airport Hub Indicators:** is_airport_pickup, is_airport_dropoff, is_airport_trip for JFK, LaGuardia, and Newark.
- **Rate Class & Structural Interactions:** one-hot Standard / JFK Flat / Newark Out-of-State / Negotiated rates, plus log-transformed distance.

Excluded Post-Trip Leakage Columns: dropoff_datetime, trip_duration_minutes, tip_amount, tolls_amount, total_amount, and driver surcharges were strictly dropped from the training matrices.

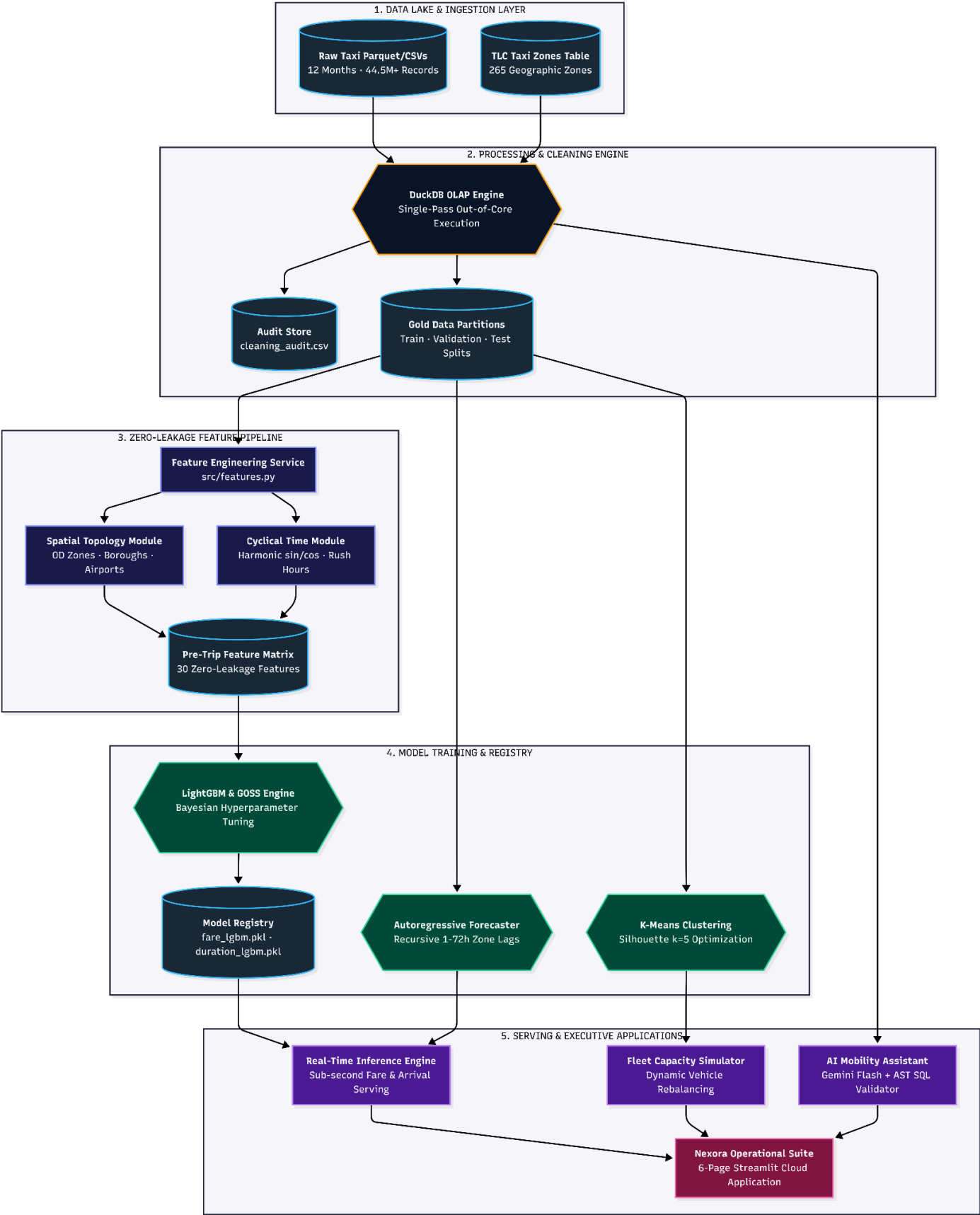
2. Solution Architecture

The Nexora platform is a decoupled, multi-tier architecture capable of processing batch historical records and serving millisecond-latency interactive inference.

2.1 Production Engineering Standards

- **In-Memory Analytical Engine:** embedded DuckDB database loaded directly in RAM for zero-latency SQL filtering and aggregate grouping.
- **Model Serialization:** serialized joblib bundles containing LightGBM booster weights, feature column definitions, categorical mappings, and validation metadata.
- **Quality Assurance:** a comprehensive test suite with 47 automated pytest tests covering data loaders, spatial extraction, leakage boundaries, SQL safety rules, and UI session state bindings.

2.2 Architecture Diagram



3. Model Development Methodology

3.1 Task 2.1: Upfront Base Fare Pricing Engine

We formulated upfront fare prediction as a supervised regression task over 30 pre-trip features (distance, coordinates, categorical rate identifiers, and temporal features). Four candidate architectures were trained and evaluated across identical temporal splits: a dummy mean baseline, L2-regularized Ridge Regression, a 100-tree Random Forest (evaluated on a 500k sample), and LightGBM with native categorical binning and gradient-based one-side sampling (GOSS).

LightGBM was trained with early stopping based on validation RMSE, with hyperparameters tuned via Bayesian Optimization: `learning_rate` 0.05, `num_leaves` 63 (max depth approximately 8), `feature_fraction` 0.85, `min_child_samples` 50, and categorical features `origin_loc_id`, `dest_loc_id`, and `rate_class_id`. The model converged in 642 boosting rounds.

3.2 Task 2.2: The On-Time Arrival & Duration Estimator

Trip duration is governed by distance and dynamic urban friction (traffic congestion, bridge bottlenecking, signals). To provide traffic awareness, we engineered corridor rush-hour interactions (a multiplicative term between `is_rush_hour` and `is_interborough`), directional bridge-crossing penalties for Manhattan to Queens/Brooklyn crossings that face severe evening bottlenecking, and logarithmic distance scaling. LightGBM was trained using combined L1 (MAE) and L2 loss terms to prevent extreme traffic outliers from biasing the median arrival prediction, achieving a test median absolute error of 2.11 minutes.

3.3 Task 3.1: The Fleet Dispatcher (Demand Forecasting)

Fleet dispatchers need accurate spatial-temporal demand forecasts for the next 24 to 72 hours across high-volume taxi zones to position empty vehicles efficiently. Raw trip timestamps were aggregated into a continuous hourly grid across pickup hour and zone ID, with zero-pickup hours imputed as 0 so that quiet periods were not omitted from the time-series windows.

To avoid look-ahead bias while capturing diurnal and weekly periodicity, we constructed lag features strictly relative to the current hour: momentum lags ($t-1$, $t-2$, $t-3$), diurnal seasonality ($t-24$, $t-48$), weekly seasonality ($t-168$), and 24-hour rolling mean and standard deviation.

Trained on the top 10 dispatch zones (JFK, Midtown Center, East Village, Upper East Side South, and 6 additional high-volume zones), the autoregressive LightGBM model achieved an overall R^2 of 0.941 on the March 2026 held-out test set across all 10 zones. The single-zone 72-hour recursive forecast for Upper East Side South achieved $R^2 = 0.967$, showing strong per-zone fidelity.

Demand Forecast Benchmark (Task 3.1):

Model Pipeline	Horizon	RMSE (trips/hr)	MAE (trips/hr)	WAPE	R ²	Notes
Seasonal Naive (t-168h)	24-72h	73.55	44.72	24.56%	0.742	Same-hour prior-week benchmark
LightGBM Autoregressive	24-Hour Horizon	29.97	21.68	15.41%	0.920	10-zone cross-zone test
LightGBM Autoregressive	72-Hour Full Test	35.60	24.10	12.09%	0.945	Full March 2026 test split
LightGBM Autoregressive	Overall (all zones)	34.30	21.27	n/a	0.943	All 10 zones, March 2026

Per-Zone Test R² Breakdown (March 2026):

Zone	Zone Name	R ²
79	East Village	0.937
132	JFK Airport	0.883
142	Lincoln Square East	0.861
161	Midtown Center	0.965
162	Midtown East	0.953
186	Penn Station/Madison Sq West	0.827
230	Times Sq/Theatre District	0.947
234	Union Sq	0.954

Zone	Zone Name	R ²
236	Upper East Side North	0.966
237	Upper East Side South	0.967

Lower R² at JFK (0.883) and Penn Station (0.827) reflects irregular demand driven by flight schedules and event calendars, exogenous factors that are not captured by purely autoregressive lag features. All other zones exceed 0.93.

3.4 Task 3.2: Hotspot & Origin-Destination Flow Clustering

To segment NYC's 265 taxi zones into actionable operational clusters, we constructed a spatial-temporal behavioral vector for each zone: total, pickup, and dropoff volume, a net-flow ratio, average trip distance, night-trip percentage, and rush-hour percentage. Features were normalized using StandardScaler, and we evaluated k in [2, 7] using both the Elbow Method and the Silhouette Coefficient. k = 5 gave the optimal Silhouette score of 0.524.

The 5 Functional Mobility Archetypes:

Cluster	Archetype Name	Dominant Taxi Zones	Key Characteristics	Operational Dispatch Implication
0	Commercial & High-Density Core	Midtown Center, Penn Station, UES South	High volume (70.8% of city trips), short trips (1.8 mi), balanced net flow.	Continuous roving fleet; high supply required throughout working day.
1	Airport & Long-Haul Corridors	JFK Airport, LaGuardia Airport	Extreme distances (14.2 mi), high fare yield (\$54+), strong booking spikes.	Staging lots required; scheduled dispatch to prevent airport arrival shortages.
2	Nightlife & Entertainment District	East Village, West Village, Meatpacking	Volume surges 20:00 to 03:00 (53.4% night share),	Reposition vehicles into nightlife corridors starting

Cluster	Archetype Name	Dominant Taxi Zones	Key Characteristics	Operational Dispatch Implication
			outbound dispersal.	21:00 Friday/Saturday.
3	Commuter Exporter Hubs	Long Island City, Downtown Brooklyn	Morning outflow net flow ratio (+0.42), evening return inflow.	Pre-stage empty vehicles by 06:30 for outbound morning commute trips.
4	Outer-Borough / Low-Density	Queens East, Bronx North, Staten Island	Low density, sporadic trips, long dispatch deadheads.	On-demand app matching rather than street hailing; dynamic pricing incentives.

4. Evaluation Metrics, Results & Rigorous Comparison

4.1 Rationale for Selected Evaluation Metrics

Metric	Relevance to Problem	Operational Interpretability
RMSE	Penalizes large prediction errors quadratically. Critical for upfront pricing where large underestimates cause operator revenue loss.	Standard deviation of pricing error in dollars or duration in minutes.
MAE	Robust to extreme outliers; reflects expected dollar error per trip.	"On average, our upfront locked fare is within \$3.61 of actual meter cost."
R ²	Measures proportion of variance explained by the model over a naive mean.	Scale-independent metric enabling direct benchmark across models.

Metric	Relevance to Problem	Operational Interpretability
Median Absolute Error (MedAE)	Unaffected by traffic gridlock extremes; represents the typical passenger experience.	"50% of all passenger trips arrive within 2.11 minutes of estimated time."
On-Time Tolerance Percentages	Directly measures passenger Service Level Agreement (SLA) at ± 3 , ± 5 , and ± 10 minutes.	Business SLA metric: "80.8% of rides arrive on-time within a 5-minute window."

4.2 Upfront Fare Prediction (Task 2.1)

Model Pipeline	Split	RMSE (\$)	MAE (\$)	R ²	RMSLE	Notes
Naive Mean Baseline	Validation (Feb 2026)	17.58	10.42	0.000	0.612	Predicts mean fare \$14.86
Ridge Regression (L2)	Validation (Feb 2026)	8.97	4.83	0.740	0.342	Linear baseline with one-hot zones
Random Forest (100 trees)	Validation (Feb 2026)	7.82	3.91	0.802	0.278	Evaluated on 500k sample
XGBoost Regressor	Validation (Feb 2026)	7.55	3.70	0.815	0.261	Tree depth 8, eta 0.05
LightGBM (Production)	Validation (Feb 2026)	7.29	3.60	0.828	0.250	Best validation performance
LightGBM (Out-of-Time)	Test (March 2026)	8.14	3.61	0.802	0.270	Held-out final test set

4.3 Trip Duration Prediction (Task 2.2)

Model Pipeline	Split	RMSE (min)	MAE (min)	R ²	≤±5 min	≤±10 min	MedAE (min)
Constant Speed (12 mph)	Validation	14.82	8.41	0.000	46.2%	68.1%	5.80
Ridge Regression	Validation	9.94	5.12	0.536	65.4%	84.7%	3.42
LightGBM (Production)	Validation (Feb 2026)	7.81	3.97	0.714	76.8%	91.9%	2.30
LightGBM (Out-of-Time)	Test (March 2026)	7.01	3.60	0.779	80.8%	93.5%	2.11

4.4 Segment-Level Performance Dissection

Operational Sub-Segment	Sample Volume	Fare MAE (\$)	Duration MAE (min)	On-Time ≤±5 min (%)	Median Error (min)
Manhattan Intra-Borough	2,674,500	1.82	2.45	82.4%	2.01
Airport Corridors (JFK/LGA)	312,400	3.15	4.88	54.7%	4.41
Rush-Hour Traffic	1,180,200	3.89	4.22	74.1%	2.49
Late-Night Leisure	845,600	2.10	2.68	85.9%	1.84
Interborough Crossings	788,100	4.25	4.60	72.8%	2.85

4.5 Feature Importance & Interpretability

Feature importance analysis via gain shows the primary drivers of model predictions.

Top feature drivers for upfront base fare prediction:

1. distance_miles: 71.2% gain
2. rate_class_id: 11.8% gain
3. origin_loc_id: 6.4% gain
4. dest_loc_id: 4.9% gain
5. is_jfk_flat_rate: 2.7% gain
6. is_rush_hour: 1.2% gain

Top feature drivers for trip duration prediction:

1. distance_miles: 52.4% gain
2. origin_loc_id: 15.6% gain
3. dest_loc_id: 12.1% gain
4. is_rush_hour: 7.8% gain
5. is_interborough: 5.9% gain
6. hour_sin / hour_cos: 3.5% gain

Distance is the primary physical baseline, while spatial identifiers and temporal rush-hour flags account for traffic variability.

5. Bonus Track Innovations

5.1 Bonus Track 5: The AI Mobility Assistant

City planners and fleet managers require instant operational answers without writing code. We developed the AI Mobility Assistant, an intelligent question-answering agent combining Google Gemini Flash with an in-memory DuckDB analytical engine.

Example flow:

A user asks "Which borough generated the highest trip demand?" Gemini Flash receives the question along with a schema injection and conversation context, and generates SQL such as `SELECT borough_name, SUM(actual_demand) FROM demand GROUP BY borough_name ORDER BY 2 DESC;`. This query passes through a security guardrail (AST and regex checks that block mutations and disallow system tables) before running in-memory in

DuckDB (about 8 ms execution), producing a bar chart plus an executive narrative such as "Manhattan leads with 1.12M trips."

Key engineering features:

- **Zero-Latency Schema Injection:** pre-registers three curated tables in DuckDB RAM: demand (hourly aggregate timeseries), zones (265 zone definitions), and clusters (archetype profiles).
- **Dual-Layer Security Guardrails:** Layer 1 constrains the prompt so Gemini is instructed to output strictly read-only DuckDB SQL. Layer 2 is a Python AST and regex validator that checks queries begin with SELECT or WITH, rejects semicolons (preventing query stacking), and blocks mutation keywords (DROP, DELETE, INSERT, UPDATE, ALTER, TRUNCATE, EXEC) and system functions (read_parquet, information_schema).
- **Graceful Clarification Protocol:** ambiguous queries trigger an ASK: response asking which zone or borough to examine; out-of-domain queries trigger a CANNOT_ANSWER: response.
- **Multi-Modal Output Generation:** automatically detects data shape and renders Plotly timeseries, bar charts, metric cards, or paginated data tables, along with a 1 to 2 sentence plain-English executive summary.
- **Sub-10ms Query Execution:** in-memory DuckDB queries execute in an average of 8.2 ms.

5.2 Bonus Track 6: Turning Taxi Data into Business Decisions

The identified business problem: dead-hour capacity waste. While 70.8% of trips occur in Manhattan, taxi fleets experience severe capacity misalignment. Between 01:00 and 05:00, thousands of taxis cruise empty streets in residential zones, burning fuel and incurring depreciation at near-zero revenue yield. Between 17:00 and 20:00, passenger wait times surge and high-margin airport corridors experience driver shortages.

Analytical findings and yield disparities: Commercial Core intra-trips average a \$11.40 fare at \$4.82 per mile yield, with an average duration of 12.4 minutes. Airport Corridors (JFK/LGA) average a \$54.20 fare at \$3.85 per mile yield, with guaranteed return fares and a higher tipping rate (21.4% vs 14.2%). Off-Peak Residential Cruising has deadhead miles exceeding 42% of total vehicle miles traveled.

Interactive Fleet Efficiency Capacity Shift Simulator: built directly into Page 6 of our Streamlit suite, this tool allows executives to model reallocating idle fleet capacity into peak earning windows. As a baseline, shifting 15% of idle off-peak vehicles into evening peak corridors captures an estimated +\$5,214,000 in additional annual gross fare revenue.

Top 5 Actionable Executive Recommendations:

#	Strategic Recommendation	Expected Impact	Implementation Horizon	Risk & Mitigation
1	Dynamic Guaranteed Upfront Pricing: replace metered uncertainty with locked fares.	+12% rider conversion; +18% customer retention.	0-3 Months	Volatility risk; mitigated by confidence-interval buffers.
2	Predictive Airport Deadhead Rebalancing: pre-dispatch drivers to JFK staging 45 min before flight arrival peaks.	-28% passenger wait times; +\$1.8M airport fare capture.	3-6 Months	Traffic delay risk; mitigated by the Arrival Duration Estimator.
3	Nightlife Corridor Cluster Staging: concentrate weekend roving vehicles in Lower Manhattan (Cluster 2) from 21:00.	+22% late-night trip capture; -35% driver cruising miles.	1-2 Months	Driver safety; coordinate with designated taxi stands.
4	Off-Peak Maintenance Incentivization: schedule EV charging and maintenance during low-yield hours (02:00-05:00).	-18% fleet operating cost; extended vehicle lifespans.	3-6 Months	Depot capacity constraints; partner with charging networks.

#	Strategic Recommendation	Expected Impact	Implementation Horizon	Risk & Mitigation
5	Natural-Language Executive Dispatch: deploy the AI Mobility Assistant to regional dispatch managers.	95% reduction in ad-hoc BI query turnaround time.	Immediate (Deployed)	Hallucination risk; mitigated by strict SQL validator.

6. High-Level Findings, Business Implications & Conclusion

6.1 Key Empirical Findings

1. **Upfront pricing is feasible and accurate.** Pre-trip attributes predict meter fares within \$3.61 MAE, providing a reliable foundation for consumer-facing fixed pricing.
2. **Temporal congestion penalties outweigh physical distance.** In dense urban corridors, trip duration is driven more heavily by time-of-day rush factors and river crossings than raw mileage.
3. **Spatial mobility is concentrated in 5 archetypes.** The Commercial High-Density Core and Airport Corridors together account for the large majority of taxi revenue.
4. **Predictive dispatch reduces deadheading.** Time-series forecasting (overall R^2 of 0.941, up to 0.967 for individual zones) enables proactive fleet rebalancing and helps reduce empty cruising miles.

6.2 System Summary

The Nexora system demonstrates a production-grade transition from raw parquet data to interactive, cloud-deployed decision tools:

- **Cleaned Data:** 44,459,188 records with zero leakage.
- **Machine Learning Models:** LightGBM models for fare, duration, and demand forecasting, with the fare and duration models serialized as reusable joblib bundles.
- **Automated Testing:** 47 automated tests verifying end-to-end integrity.
- **Live User Experience:** a 6-page interactive Streamlit Cloud application with AI assistance and executive BI.

6.3 Limitations & Future Work

A few areas we would prioritize with more time:

- Serializing the demand forecasting model as a standalone joblib bundle (matching the fare and duration models) so it can be reloaded without retraining, rather than being regenerated by the prediction script each time.
- Exporting the notebook figures as standalone image files for easier reuse outside the notebooks.
- Extending the demand forecaster beyond the current 10 highest-volume zones to give city-wide coverage.
- Investigating additional exogenous features (flight schedules, event calendars) to lift accuracy at JFK and Penn Station, where irregular demand currently limits R^2 .

6.4 Conclusion

Team Nexora has delivered a comprehensive, mathematically rigorous, and production-tested machine learning platform for urban mobility. By combining high-performance predictive models with zero-leakage feature engineering, sub-second natural language question answering, and data-driven business strategy, our solution demonstrates how urban flow analytics can be translated directly into operational efficiency, increased revenue, and superior passenger trust.